llmXive Automated Review of RoboDojo: A Unified Sim-and-Real Benchmark for Comprehensive Evaluation of Generalist Robot Manipulation Policies

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the llmXive agent-prompt library; each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (8 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, and Statistical Analysis. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper contains numerous citations to models and benchmarks that are either missing from the provided bibliography or have mismatched citation keys. Specifically, the text cites `rdt2`, `spiritspirit`, `cai2026internvla`, `jiang2025galaxea`, `cai2026xiaomi`, `zheng2025x`, `bjorck2025gr00t`, `black2024pi_0`, `ye2026starvla`, `spiritspirit`, `yu2026dm0`, `li2025spatial`, and `lyu2026lda`, but these keys are absent from the reference list. Additionally, there are mis-attributions, such as `li2026causalworldmodelingrobot` being cited for "LingBot-VA" when the bibliography entry is for "WALL-WM". These issues undermine the credibility of the leaderboard results and the claims about the state of the art. The authors must verify and correct all citation keys and ensure that every cited model has a corresponding, accurate entry in the bibliography. This is a critical issue for a benchmark paper, as the validity of the comparisons depends on the accuracy of the references.

Action items.

- [**writing**] The paper contains numerous citations to models and benchmarks that are either missing from the provided bibliography or have mismatched citation keys. Specifically, the text cites rdt2, spiritspirit, cai2026internvla, jiang2025galaxea, cai2026xiaomi, zheng2025x, bjorck2025gr00t, black2024pi_0, ye2026starvla, spiritspirit, yu2026dm0, li2025spatial, and lyu2026lda, but these keys are absent from the reference list. Additionally, there are mis-attributions, such as li2026causalworldmodelingrobot b

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_figure_critic.md`

Figure 1

Figure 1 consists of a collage of university logos but lacks a descriptive caption and any data visualization elements, rendering its purpose and scientific contribution unclear.

Figure 2

The figure provides a clear and comprehensive visual overview of the RoboDojo benchmark components as described in the caption. However, the leaderboard panel displays a future date (2026), which is likely a placeholder error.

Figure 3

The figure effectively visualizes the task overview with clear images and labels, but the caption contains editing artifacts and fails to explain the specific 'DLC' and 'Domain Randomization' diagrams shown in the top right.

Figure 4

The figure effectively visualizes the task diversity of the RoboDojo benchmark across simulation and real-world settings. However, the caption fails to describe the significant 'Domain Randomization' section shown in the top right, and the use of large, stylized titles detracts from the professional presentation.

Figure 5

The figure effectively visualizes a diverse set of robot manipulation tasks, but the caption's claim of 24 skills is contradicted by the visual content, which repeats the 'Pick' label and omits 'Pulling'.

Figure 6

Figure 6 effectively displays representative key frames of real-world manipulation tasks as described in the caption. The image is clear, and the visual content aligns with the claim of showing challenging tasks across different robot embodiments.

Figure 7

Figure 7 effectively illustrates the four future extension scenarios (Dexterous, Humanoid, Tactile, and Mobile Manipulation) with clear, labeled images that align perfectly with the caption's description. The visual presentation is uncluttered and successfully communicates the intended roadmap for the RoboDojo benchmark.

Figure 8

The figure clearly displays the hardware components with appropriate labels, but the caption contains a formatting artifact ('Blue_1') and a potential discrepancy regarding the quantity of cameras shown versus described.

Figure 9

The figure is rendered as a solid black image, making it impossible to verify the caption's claims regarding domain randomization effects such as lighting or clutter variations.

Figure 10

Figure 10 is a clear and well-annotated schematic of the RoboDojo-RealEval hardware platform. The visual components (cameras, robots, lighting, workstation) are explicitly labeled and align perfectly with the caption's description of the system's standardized features.

Figure 11

The figure displays a diverse collection of 3D assets but fails to visually demonstrate the 'articulated' or 'deformable' categories claimed in the caption, and lacks labels to map specific objects to these categories.

Action items.

- [**writing**] Figure 1: The caption is missing; it currently reads '(no caption) [university_logo.pdf]', which is a placeholder rather than a descriptive summary of the figure's content.
- [**science**] Figure 1: The figure displays a collection of university logos without any axes, data, or visual comparison metrics, making it unclear what scientific claim or relationship this figure is intended to support.
- [**writing**] Figure 2: The 'Leaderboard' panel displays a date of '2026 Jul 1', which is a future date relative to the current time, suggesting a placeholder or error in the rendered graphic.
- [**writing**] Figure 2: The 'Leaderboard' panel contains a footer 'maintained by the AI MMLAB Club' which appears to be a specific attribution that may need verification or removal if not applicable to the final publication.
- [**writing**] Figure 3: The caption text contains a stray artifact 'Blue_1' at the beginning and ends abruptly with 'diagnosis of policy per', indicating incomplete editing.
- [**writing**] Figure 3: The top-right section includes a 'DLC (train-only)' label and a 'Normal Train+Eval vs Random Eval' diagram that are not explained in the caption, leaving their purpose ambiguous.
- [**writing**] Figure 4: The image contains a large 'Domain Randomization' section with a legend and 'DLC (train-only)' panel that are not mentioned in the caption, creating a disconnect between the visual content and the text description.
- [**writing**] Figure 4: The title 'Simulation Benchmark' and the 'Real-World Benchmark' section header are rendered in a large, stylized font that is inconsistent with standard scientific figure labeling conventions.
- [**writing**] Figure 5: The caption claims to cover 24 manipulation skills, but the figure displays

24 images with only 23 unique labels (the label 'Pick' appears twice in the first row, while 'Pulling' is missing).

- [**writing**] Figure 5: The label 'HandOver' uses inconsistent CamelCase capitalization compared to the other labels which are either Title Case (e.g., 'HandOver' vs 'Place') or lowercase.
- [**writing**] Figure 8: The caption contains the artifact 'Blue_1' at the beginning of the title, which appears to be a formatting error or leftover placeholder.
- [**writing**] Figure 8: The caption lists '(a) two Gemini 305 wrist cameras', but the image shows only a single camera unit labeled (a); the text should clarify if this represents a pair or correct the count.
- [**writing**] Figure 9: The caption claims to visualize variations in background, lighting, clutter, and object appearance, but the rendered image is a solid black rectangle with no visible content to verify these claims.
- [**science**] Figure 11: The caption claims to show 'articulated' and 'deformable' assets, but the image displays only static, rigid objects (e.g., toasters, shirts, figurines) with no visible joints, articulation, or deformation states to substantiate these categories.
- [**writing**] Figure 11: The image lacks any labels, legends, or visual indicators to distinguish which specific assets belong to the 'rigid', 'articulated', or 'deformable' categories mentioned in the caption.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

This paper presents a comprehensive new benchmark for robot manipulation. The writing is generally clear, but suffers from a common issue in specialized fields: assuming the reader is already familiar with certain acronyms and shorthand. Several acronyms (SOTA, DLC, RAG, XPolicyLab, HP) are used without being defined at first use, which creates a barrier to comprehension for a reader from an adjacent field (e.g., computer vision, general machine learning). While these terms may be well-known within the robotics community, they are not universally understood.

Additionally, the frequent use of "task" to refer specifically to a RoboDojo task could be clarified with a brief explanation on first use.

Addressing these minor issues will significantly improve the accessibility of the paper to a broader audience within the machine learning community. The technical content is strong, and the benchmark itself appears to be a valuable contribution.

Action items.

- **[writing]** Section 1.1 uses 'SOTA' without defining it at first use. Expand to 'state-of-the-art' on first mention.
- **[writing]** Section 2.1 uses 'DLC' without defining it at first use. Define as 'demonstration learning collection' or similar.
- **[writing]** Section 3.2 uses 'RAG' without defining it at first use. Expand to 'retrieval-augmented generation'.
- **[writing]** Section 3.2 uses 'XPolicyLab' without defining it at first use. Briefly explain it is a unified policy interface.
- **[writing]** Section 4.1 uses 'HP' without defining it at first use. Define as 'Heterogeneous Parallelization'.
- **[writing]** Throughout the paper, 'task' is used in a very specific sense (a RoboDojo task). While not strictly jargon, consider adding a clarifying phrase on first use (e.g., 'RoboDojo task').

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

The paper presents a coherent argument for the RoboDojo benchmark, with clear premises regarding the limitations of existing benchmarks and logical conclusions about the necessity of a unified sim-and-real evaluation system. The structure generally holds, with the experimental results supporting the stated claims about policy limitations.

However, there are three specific instances where the internal consistency of the argument or the alignment between text and data is slightly broken:

1. **Numerical Discrepancy in Task Count:** In Section 4.1.2 ("Training Data Setting"), the text states there are "**35 task directories**" used for training. However, the breakdown in Section 4.1.1 lists 12 (Generalization) + 6 (Memory) + 8 (Long-Horizon) + 8 (Precision) = 34 tasks, as the "Open" dimension (8 tasks) is explicitly excluded from training. The sum of the components (34) contradicts the stated total (35). This creates a logical gap in the dataset description. The authors must verify if one task is split across directories or if the count is a typo.
2. **Ambiguous Causal Phrasing:** In Section 5.1, Finding 2, the text claims: "Spatial Forcing reduces the drop from 72.2% to 67.2%." This phrasing logically implies that the Spatial Forcing policy actively reduced a drop that was previously 72.2% (attributed to another policy, $\pi_{0.5}$). The data actually shows two independent measurements: $\pi_{0.5}$ has a 72.2% drop, and Spatial Forcing has a 67.2% drop. The current phrasing conflates a comparison

with a causal intervention. The argument should be rephrased to state that Spatial Forcing *exhibits* a lower drop (67.2%) compared to $\pi_{0.5}$ (72.2%), removing the implication of active reduction.

3. **Disconnected Data Point:** In Section 5.2.1, the text focuses on the internal efficiency gain of RoboDojo (77.4 vs 40.0 interactions/s). However, Table 1 includes a row for "RoboTwin 2.0" (44.6 interactions/s) without a corresponding textual comparison in the paragraph. This leaves the inclusion of the third-party benchmark data point logically unanchored in the immediate argument. A brief sentence explicitly comparing RoboDojo's non-heterogeneous baseline to RoboTwin 2.0 would close this gap.

Addressing these points will ensure the paper's internal logic is watertight and the data presentation aligns perfectly with the narrative.

Action items.

- [**writing**] Section 4.1.2 claims "35 task directories" for training, but Section 4.1.1 lists 12+6+8+8=34 training tasks (Open excluded). Verify if a task is split or correct the count to 34.
- [**writing**] Section 5.1 Finding 2 states "Spatial Forcing reduces the drop from 72.2% to 67.2%," implying causality between policies. Rephrase to clarify Spatial Forcing *achieves* a 67.2% drop, compared to 72.2% for pi_0.5.
- [**writing**] Table 1 includes RoboTwin 2.0 (44.6 interactions/s) but Section 5.2.1 only compares RoboDojo's internal modes. Add a sentence explicitly comparing RoboDojo's non-heterogeneous mode to RoboTwin 2.0 to justify the table entry.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The paper makes several claims regarding the scope and generality of the RoboDojo benchmark that exceed the specific experimental evidence provided.

First, the **Title** ("Comprehensive Evaluation of Generalist Robot Manipulation Policies") and the **Abstract** imply a universal applicability to "generalist" policies across the field. However, the **Real-World Benchmark** (Section 4.2) is strictly limited to three specific bimanual robot embodiments (ARX X5, Piper, Piper X). The **Future Extensions** section (Section 6) explicitly lists dexterous hands, humanoids, and mobile manipulation as *future* work, confirming they are not part of the current evaluation. By using the unqualified term "Generalist" and "Comprehensive" without qualifying the embodiment constraints, the paper overstates the current scope of its validation. The benchmark is comprehensive *for the tested bimanual domain*, but not for generalist manipulation in the broader sense of all robotic forms.

Second, the **Conclusion** asserts that the benchmark provides a testbed for "omni-manipulation policies." This term suggests a capability covering all forms of manipulation (dexterous, mobile, whole-body). Since the current results (Section 5) are exclusively on bimanual arms and the other modalities are listed as future work, this claim is a projection rather than a demonstrated fact. The language should be adjusted to reflect that the benchmark is a *foundation* for omni-manipulation evaluation, not the current realization of it.

Third, in **Section 5.1 (Finding 2)**, the statement that "Scene-level randomization causes broad performance collapse" is slightly overgeneralized. While the data shows a massive drop for Hy-Embodied-0.5-VLA (92.9%), other policies like RDT-1B show a smaller relative drop (64.6%). While the trend is negative, "collapse" implies a total failure across the board, which the data shows is not uniform. A more precise phrasing would acknowledge the severity while noting the variance across architectures.

These issues are primarily matters of **scope and generalization**. The evidence is strong for the specific domain tested (bimanual, specific simulators, specific tasks), but the rhetoric extends this to the entire field of generalist/omni-manipulation without the necessary qualifiers. Narrowing the claims to match the specific experimental boundaries (bimanual, specific embodiments) will align the rhetoric with the evidence.

Action items.

- [**writing**] Title claims 'Comprehensive Evaluation of Generalist Robot Manipulation Policies,' but the benchmark is restricted to three specific bimanual embodiments (ARX X5, Piper, Piper X) and excludes dexterous hands, humanoids, and mobile bases (Section 6, Fig 5). The title implies a broader scope than the evidence supports. Narrow the title to 'Bimanual' or add 'Bimanual' to the scope description in the abstract.

- [**writing**] The abstract states RoboDojo provides a 'unified sim-and-real benchmark' for 'generalist' policies, yet the real-world evaluation (Section 4.2) is limited to 18 tasks across only 3 specific robot models. The term 'generalist' in the title and abstract suggests applicability across arbitrary embodiments, which is not tested. Qualify the claim to 'bimanual generalist policies' or explicitly state the embodiment limitation in the abstract.

- [**writing**] The conclusion claims the benchmark tracks progress toward 'omni-manipulation policies,' but current results (Section 5) cover only bimanual arms. 'Omni-manipulation' implies dexterous/mobile/whole-body capabilities listed as 'Future Extensions' (Section 6) and not yet demonstrated. Remove 'omni-manipulation' or rephrase to reflect the benchmark is a step toward, not a realization of, that scope.

- [**writing**] Section 5.1 Finding 2 claims 'Scene-level randomization causes broad performance collapse,' citing a 92.9% drop for one model. While data supports a significant drop, 'broad performance collapse' implies a universal failure mode. Table 1 (e002) shows some policies (e.g., RDT-1B) have lower relative drops (64.6%). Hedge the claim to 'causes severe performance degradation in most evaluated policies' to reflect data variance.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_safety_ethics.md`

The paper presents a benchmark and evaluation framework for robot manipulation policies. From a safety and ethics perspective, the work is low-risk. The primary data sources are synthetic simulation environments (Isaac Sim) and real-world teleoperation demonstrations collected by the authors. The paper explicitly states that real-world demonstrations were collected via “homogeneous leader-follower teleoperation” by “four different operators” (Appendix, Section “Real-World Training Data Details”). While the paper does not contain a formal IRB statement, the nature of the data (robotic manipulation trajectories, not sensitive personal health or private communications) and the context of standard robotics research typically allow for exemption or minimal risk classification. The authors describe a manual filtering process for data quality but do not mention collecting PII, which is consistent with the task descriptions (e.g., “stack bowls,” “insert tubes”).

The paper addresses safety in the context of system stability and hardware protection. Section “Real-World Evaluation Details” notes that the evaluation manager may “manually stop a trial if the robot exhibits unsafe behavior that could damage the platform,” and the platform includes an “emergency stop function.” This is an appropriate mitigation for the physical risks inherent in real-world robot evaluation. The paper does not propose dual-use capabilities for harm (e.g., autonomous weaponization, surveillance, or biological synthesis), nor does it release datasets containing re-identifiable information. The “Leaderboard Governance” section clarifies the non-profit, academic nature of the evaluation, mitigating concerns about undisclosed commercial conflicts of interest driving the results.

No specific, non-trivial safety or ethical risks were identified that require disclosure or mitigation beyond what is already present. The work falls squarely within the norms of standard robotics benchmarking research.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a comprehensive benchmark, but the evidentiary strength of the comparative claims in the leaderboards is weakened by insufficient reporting of variance and experimental replication.

First, the Simulation Leaderboard (Table 1) presents aggregate success rates and scores as single point estimates (e.g., 8.80% for Hy-Embodied-0.5-VLA). While the text mentions evaluation over 3 seeds, the table does not report the standard deviation or confidence intervals for these means. In robotics benchmarks, performance can vary significantly across seeds due to stochastic initialization and environment dynamics. The Appendix (Table 4) reveals high per-task variance (e.g., a 23.1% standard deviation in success rate for the 'store_in_safe' task for $\pi_{0.5}$). Without reporting this variance in the main leaderboard, it is impossible to determine if the observed differences between policies (e.g., 8.80% vs 8.04%) are statistically significant or merely within the noise of the evaluation process. The authors should report mean ± standard deviation for all leaderboard entries.

Second, the claim that "scene-level randomization causes broad performance collapse" (Finding 2, Section 5.1) relies on a comparison between "Standard" and "Random" settings. However, the experimental design appears to compare a single "Standard" run against a single "Random" run (or a small set) without explicitly controlling for the random seed used in the Standard condition. If the "Standard" set happened to be easier or the "Random" set harder due to a specific seed, the magnitude of the "collapse" could be exaggerated. To robustly support this claim, the authors should report performance across multiple seeds for both conditions or demonstrate that the drop is consistent regardless of the specific random seed used for the "Standard" baseline.

Third, the Real-World Benchmark results (Section 5.2) are based on a single seed per embodiment (Section 4.2). While real-world evaluation is expensive, relying on a single seed with only 10 trials per task makes the comparative ranking of policies highly susceptible to luck or specific initial conditions. The paper claims to reveal "execution instability," but the current design cannot distinguish between a policy that is generally unstable and one that simply encountered a rare failure mode in a single run. To support the comparative claims in the real-world leaderboard, the authors should ideally report results across at least 3 seeds, or at minimum, explicitly acknowledge that the rankings are preliminary and sensitive to the specific seed chosen.

These issues do not invalidate the benchmark itself, but they prevent the specific performance claims and rankings from being fully trusted as generalizable scientific evidence.

Action items.

- **[writing]** The paper presents a comprehensive benchmark, but the evidentiary strength of the comparative claims in the leaderboards is weakened by insufficient reporting of variance and experimental replication. First, the Simulation Leaderboard (Table 1) presents aggregate success rates and scores as single point estimates (e.g., 8.80% for Hy-Embodied-0.5-VLA). While the text mentions evaluation over 3 seeds, the table does not report the standard deviation or confidence intervals for these means. In robo

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The statistical reporting in the RoboDojo paper is generally transparent regarding the number of seeds and episodes, but it lacks necessary uncertainty quantification for its primary quantitative claims.

Missing Uncertainty in Leaderboards: The core contribution of the paper is the benchmark leaderboard (Tables 1 and 2), which ranks 30 policies. These tables report point estimates (e.g., "8.80% success rate") to two decimal places. In stochastic environments like robot manipulation, a difference of 0.01% is meaningless without a measure of variance. While the authors report standard deviations for a *subset* of policies in Section 5.3 (Stability) and in the appendix tables for specific policies (e.g., `part:pi_05`), the main summary tables do not include standard deviations (SD) or confidence intervals (CI) for the aggregate scores. This prevents readers from determining if the ranking differences (e.g., between 8.80% and 8.04%) are statistically significant or within the noise of the evaluation. The field norm for such benchmarks is to report "Mean ± SD" over the reported seeds.

Unverified Significance Claims: In Section 5.1, Finding 2, the authors state that scene randomization causes a "broad performance collapse," citing a "92.9% relative drop" for one policy. This is a comparative claim derived from a single aggregate number (Standard Score vs. Random Score). Without a reported p-value, standard error of the difference, or a confidence interval for the drop, the claim of a "collapse" is descriptive rather than inferential. Given the large number of episodes (2,100 total), the variance might be low, but the paper must explicitly state the statistical test used (e.g., paired t-test across tasks) to validate that the drop is not a random fluctuation.

Improper Baseline for Efficiency Claims: Table 3 compares the evaluation throughput of RoboDojo against RoboTwin 2.0. The caption notes that RoboDojo runs on "8xRTX 4090" while RoboTwin 2.0 runs on "two simulation processes per GPU" (hardware unspecified, likely different). Comparing raw throughput (interactions/s) across different hardware configurations and process parallelization strategies is a methodological error. The "1.94x speedup" is only valid if the baselines are normalized to the same hardware and process count. As written, the comparison conflates algorithmic efficiency with hardware capability, rendering the quantitative claim of "substantial improvement" unsupported.

Action items.

- [**writing**] Tables 1 and 2 report mean success rates to two decimals without SD or CI. Report mean ± SD for all 30 policies in the leaderboards to allow assessment of ranking significance.
- [**writing**] Section 5.1 Finding 2 claims a '92.9% relative drop' without a statistical test or standard error. Add a paired test or SE for the difference to support the 'collapse' claim.
- [**science**] Table 3 compares throughput on different hardware/process settings. Re-calculate speedup against a baseline run on identical 8xRTX 4090 hardware or explicitly state the mismatch.